

3 Cholera is an infectious disease caused by the bacterium *Vibrio cholerae*.

- (a) A person who lives in an area where cholera is endemic (always present), and who does **not** have access to vaccines, can use different ways to prevent infection by *V. cholerae*.

Outline the ways in which a person who does **not** have access to vaccines can prevent infection by *V. cholerae*.

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- (b) In 2023, the highest number of cases of cholera and deaths from cholera were in the World Health Organization Africa region.

Table 3.1 shows data for the five countries with the highest number of cases of cholera in the Africa region. The case fatality rate (CFR) is the number of deaths expressed as a percentage of the number of cases. This has been included for three countries.

Table 3.1

country	total number of cases	total number of deaths	CFR /%
Democratic Republic of the Congo	52 654	473	
Malawi	32 530	817	2.5
Ethiopia	30 389	446	1.5
Somalia	18 304	46	0.3
Sudan	8 895	246	

- (i) A student made a statement after studying Table 3.1.

As the number of cases of cholera decreases, the CFR also decreases.

Explain whether you agree with this statement.

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- (ii) The treatment for cholera varies depending on the severity of the disease. With the correct treatment, a person with cholera can make a full recovery.

The CFR for a particular country indicates if any improvements can be made in the response to the disease, for example in the availability and quality of treatment.

Table 3.1 shows that Malawi and Ethiopia have approximately the same number of cases of cholera but very different CFRs.

Suggest explanations for the differences in the CFR between Malawi and Ethiopia.

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- (c) Genes that are essential for bacteria to survive are located on a large, circular DNA molecule in the cell cytoplasm. *V. cholerae* is unusual as it has two of these circular molecules.

- (i) Explain what is meant by a gene.

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- (ii) *V. cholerae* is a unicellular organism with a diameter similar to other bacteria.

V. cholerae has structural features that provide evidence that it is a bacterium, such as the presence of circular, double-stranded DNA.

State **other** structural features present in a *V. cholerae* cell, which would provide evidence that the organism is a bacterium.

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